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Thursday Commissioning Simulation — Instructor Pack

ORBIT NORTH · Thu 20 August 2026 · uOttawa Cyberrange

ORBIT NORTH
Summer 2026

Faculty of Engineering · School of Engineering Design and Teaching Innovation (SEDTI)

GNG 3100 · SYS 5186 · ORBIT NORTH · Issue 1.0 · 2026-10-28

Classification: Internal

Thursday Commissioning Simulation — Instructor Pack

Confidential — instructor and simulator operator only.

This pack covers the Thursday simulation day. Teams swap operator/ observer roles relative to Wednesday. Day target is **First Light** on the multispectral imager.

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1. Day at a glance

When	What	Mode
09:00 – 09:30	Standup, scenario brief to both teams	Instructor-led
09:30 – 12:00	**Pass-set 1** — Team A operates two compressed passes; Team B observes	Commissioning sim
12:00 – 13:00	Lunch + observer prep	—
13:00 – 13:15	Reset + brief for Pass-set 2	Instructor-led
13:15 – 15:45	**Pass-set 2** — Team B operates two compressed passes; Team A observes	Commissioning sim
15:45 – 17:00	Joint debrief, observer presentations, anomaly tickets	Plenary + individual

Each pass-set covers ~4 simulated ground passes in 2.5 h real time using a compressed-time multiplier (3×–5× between passes, 1× during passes). The simulator handles the speed switching.

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2. Scenario configuration

- **Breakpoint:** BP-3 — Commissioning Day Start (NEW; spec in `breakpoint_spec.md`).
- **Phase at start:** 4 (LEOP-complete).
- **Initial visibility:** Iqaluit, AOS at scenario T+0.
- **Planned passes during shift:** 4 (Iqaluit, Troll, Iqaluit, Troll — actual visibility depends on orbit).
- **Speed multiplier:** instructor-controlled; defaults to 3× between passes, 1× during passes.

[Claude Code: confirm the simulator's compressed-time CLI options.]

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3. Procedure expectations

The team is expected to drive commissioning in approximately this order across the four passes:

Pass 1. COM-001 EPS Checkout · COM-002 TCS Verification · COM-005 AOCS Mode Transitions (NADIR) · COM-006 TTC Link Verification.

Pass 2. COM-007 OBDH Checkout · COM-008 FDIR Configuration (enable S12 monitoring, verify the S12/S19 rule sets).

Pass 3. COM-009 Payload Power On · COM-010 FPA Cooler Activation (this runs through the eclipse and into pass 4 — heat sink during eclipse, cool-down on the day side).

Pass 4. COM-011 Payload Calibration · **COM-012 First Light** — acquire one image of an ocean-current target listed in `planning/imaging_targets.yaml`.

End-of-shift target state: phase 5+ (NOMINAL ready), payload calibrated, one usable image acquired.

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4. Anomaly injection menu

Pick one primary and one optional secondary per team. Injection files live in `~/SpaceMissionSimulation/configs/eosat1/scenarios/`.

Severity	Scenario file	What students see	Recovery
Easy	`eps_overcurrent.yaml` (line=`payload`)	Overcurrent on payload power-on	CTG-012 Overcurrent Response
Easy	`gs_rf_degradation.yaml`	BER climbs mid-pass	CTG-014 BER Anomaly
Medium	`fpa_overtemp.yaml`	FPA cooler can't hold its −15 °C setpoint	CTG-004 Thermal Exceedance + payload safe
Medium	`payload_corrupt_image.yaml`	First image returns corrupt	CTG-015 Corrupted Image Recovery
Hard	`aocs_sensor_cascade.yaml`	Star tracker degrades; pointing budget broken	CTG-008 Star Tracker Failure → CTG-002 + defer First Light
Hard	`eps_progressive_load_shed.yaml`	Battery degradation forces progressive load-shed during cooler activation	CTG-001 Under-Voltage Load Shed
Very hard	`aocs_wheel_failure.yaml` followed by `transponder_failure.yaml`	Wheel locks late in pass 3, then transponder fails next pass — multi-anomaly response	CTG-007 → CTG-003

[Claude Code: produce one cue card per scenario, same template as LEOP day §4.]

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5. Pass operator-side criteria

Criterion	Pass-set 1 (Team A)	Pass-set 2 (Team B)
Phase advanced to COMMISSIONING (5)	✓	✓
EPS checkout complete with no faults	✓	✓

Criterion	Pass-set 1 (Team A)	Pass-set 2 (Team B)
AOCS NADIR mode achieved, pointing < 0.5°	✓	✓
FDIR rule sets enabled and verified (COM-008)	✓	✓
Payload powered ON	✓	✓
FPA cooler at its setpoint (−15 °C)	✓	✓
First Light image acquired (or formally deferred)	✓	✓
Anomaly tickets filed for all injections	✓	✓

The "formally deferred" option is genuine: if the team correctly diagnoses that pointing is insufficient and chooses to defer imaging, they pass the criterion. If they image anyway and produce garbage, they fail.

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6. Coaching during the shift

[Claude Code: copy §6 of `sim_leop_instructor.md` and adapt for commissioning specifics. Add: watch the team's *handover quality* between passes, since pass-to-pass continuity is harder than within a single pass.]

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7. Debrief

[Claude Code: same structure as `sim_leop_instructor.md` §7, with the difference that observer worksheets used today are *Team* and *Space usage* (the other two were used Wednesday).]

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8. Safety, red lines, references

[Claude Code: copy from `sim_leop_instructor.md` §§8 – 9 and add the Commissioning-day-specific files:

- `scenarios/commissioning_day_start.yaml` (NEW)
- `procedures/commissioning/*.md`
- The `mcs/role_analysis/payload_ops_role.md` (today is heavy on payload ops).]